GROUP SCHEMES OVER A FIELD HAVE A UNIQUE IRREDUCIBLE COMPONENT THROUGH THE IDENTITY, GEOMETRICALLY IRREDUCIBLE

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Allowed standard prerequisites (editorial).

- Lemma algebra-lemma-local-flat-ff (canonical tag 00HR). cited at proof lines 1020-1024. The local ring map to the base-changed group scheme is faithfully flat.
- Lemma algebra-lemma-injective-minimal-primes-in-image (canonical tag 00FK). cited at proof lines 1024-1026. Transfers the unique-minimal-prime property back to O_{G,g}.
- Lemma varieties-lemma-bijection-irreducible-components (canonical tag 038F). cited at proof lines 1035-1038. The product of two components over k-bar is irreducible.

Source reference. Original lemma 00HR: sources/stacks-chow/tags.txt, source lines 653–653. Cited at proof lines 1020-1024 to see O_{G,g} -> O_{G_K,g'} is faithfully flat; load-bearing for the reduction.

Source reference. Original lemma 00FK: sources/stacks-chow/tags.txt, source lines 574–574. Cited at proof lines 1024-1026 to transfer the unique-minimal-prime property back along the faithfully flat map; load-bearing for part (1).

Source reference. Original lemma 038F: sources/stacks-chow/tags.txt, source lines 4234–4234. Cited at proof lines 1035-1038 to see Z_1 x_k Z_2 is irreducible; load-bearing for the uniqueness of the component through e.

Lemma 0.2. *Let G be a group scheme over a field k. Then*

(1) *every local ring $\mathcal{O}_{G,g}$ of G has a unique minimal prime ideal,*
(2) *there is exactly one irreducible component Z of G passing through e, and*
(3) *Z is geometrically irreducible over k.*

Proof. For any point $g \in G$ there exists a field extension K/k and a K-valued point $g' \in G(K)$ mapping to g. If we think of g' as a K-rational point of the group scheme G_K, then we see that $\mathcal{O}_{G,g} \to \mathcal{O}_{G_K,g'}$ is a faithfully flat local ring map (as $G_K \to G$ is flat, and a local flat ring map is faithfully flat, see Algebra, Lemma 00HR). The

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result for $\mathcal{O}_{G_K,g'}$ implies the result for $\mathcal{O}_{G,g}$, see Algebra, Lemma 00FK. Hence in order to prove (1) it suffices to prove it for k-rational points g of G. In this case translation by g defines an automorphism $G \to G$ which maps e to g. Hence $\mathcal{O}_{G,g} \cong \mathcal{O}_{G,e}$. In this way we see that (2) implies (1), since irreducible components passing through e correspond one to one with minimal prime ideals of $\mathcal{O}_{G,e}$.

In order to prove (2) and (3) it suffices to prove (2) when k is algebraically closed. In this case, let Z_1, Z_2 be two irreducible components of G passing through e. Since k is algebraically closed the closed subscheme $Z_1 \times_k Z_2 \subset G \times_k G$ is irreducible too, see Varieties, Lemma 038F. Hence $m(Z_1 \times_k Z_2)$ is contained in an irreducible component of G. On the other hand it contains Z_1 and Z_2 since $m|_{e \times G} = \mathrm{id}_G$ and $m|_{G \times e} = \mathrm{id}_G$. We conclude $Z_1 = Z_2$ as desired. $\square$